

Introduction to metallic nanoparticles

Vicky V. Mody, Rodney Siwale, Ajay Singh¹, Hardik R. Mody²

Department of
Pharmaceutical Sciences,
Appalachian College of
Pharmacy, 1060 Dragon
Road, Oakwood, Virginia,
USA 246 14, ¹Department
of Radiology, UT
Southwestern Medical
Center, 5323 Harry
Hines Boulevard,
Dallas, Texas 753 90,
²Dr. L.H. Hiranandani
College of Pharmacy,
Mumbai University,
Ulhasnagar-421 003,
India

Address for correspondence:
Dr. Vicky V. Mody,
E-mail: vmody@acpharm.org

Received : 21-06-10
Review completed : 24-07-10
Accepted : 28-08-10
DOI: 10.4103/0975-7406.72127

ABSTRACT

Metallic nanoparticles have fascinated scientist for over a century and are now heavily utilized in biomedical sciences and engineering. They are a focus of interest because of their huge potential in nanotechnology. Today these materials can be synthesized and modified with various chemical functional groups which allow them to be conjugated with antibodies, ligands, and drugs of interest and thus opening a wide range of potential applications in biotechnology, magnetic separation, and preconcentration of target analytes, targeted drug delivery, and vehicles for gene and drug delivery and more importantly diagnostic imaging. Moreover, various imaging modalities have been developed over the period of time such as MRI, CT, PET, ultrasound, SERS, and optical imaging as an aid to image various disease states. These imaging modalities differ in both techniques and instrumentation and more importantly require a contrast agent with unique physiochemical properties. This led to the invention of various nanoparticulated contrast agent such as magnetic nanoparticles (Fe_3O_4), gold, and silver nanoparticles for their application in these imaging modalities. In addition, to use various imaging techniques in tandem newer multifunctional nanoshells and nanocages have been developed. Thus in this review article, we aim to provide an introduction to magnetic nanoparticles (Fe_3O_4), gold nanoparticles, nanoshells and nanocages, and silver nanoparticles followed by their synthesis, physiochemical properties, and citing some recent applications in the diagnostic imaging and therapy of cancer.

KEY WORDS: Fe_3O_4 , gold nanoparticles, iron oxide nanoparticles, metallic nanoparticles, nanocages, nanoshells, silver nanoparticles

Nanotechnology refers to the branch of science and engineering dedicated to materials, having dimensions in the order of 100th of nm or less.^[1] The term being new, but has been widely used for the development of more efficient technology. In recent years, nanotechnology has been embraced by industrial sectors due to its applications in the field of electronic storage systems,^[2] biotechnology,^[3] magnetic separation and preconcentration of target analytes, targeted drug delivery,^[4,5] and vehicles for gene and drug delivery.^[2,4-6] Consequently, with wide range of applications available, these particles have potential to make a significant impact to the society. Although new, the history of nanomaterials dates long back to 1959, when Richard P. Feynman, a physicist at Cal Tech, forecasted the advent of nanomaterials. In one of his class he said, "There is plenty of room at the bottom," and suggested that scaling down to nanolevel and starting from the bottom was the key to future technology and advancement.^[6] As the field of nanotechnology advanced, novel nanomaterials become apparent having different properties as compared to their larger counterparts. This difference in the physiochemical

properties of nanomaterials can be attributed to their high surface-to-volume ratio. Due to these unique properties, they make excellent candidate for biomedical applications as variety of biological processes occur at nanometer scales.

In general, nanoparticles used in the field of biotechnology range in particle size between 10 and 500 nm, seldom exceeding 700 nm. The nanosize of these particles allows various communications with biomolecules on the cell surfaces and within the cells in way that can be decoded and designated to various biochemical and physiochemical properties of these cells.^[7] Similarly, its potential application in drug delivery system and noninvasive imaging offered various advantages over conventional pharmaceutical agents.^[7] In an effort to utilize nanoparticles at their full throttle, it is important that the nanoparticulate systems should be stable, biocompatible, and selectively directed to specific sites in the body after systemic administration. More specific targeting systems are designed to recognize the targeted cells such as cancer cells. This can be achieved by conjugating

How to cite this article: Mody VV, Siwale R, Singh A, Mody HR. Introduction to metallic nanoparticles. J Pharm Bioall Sci 2010;4:282-9

the nanoparticle with an appropriate ligand, which has a specific binding activity with respect to the target cells. In addition, nanoparticles provide a platform to attach multiple copies of therapeutic substance on it and hence increase the concentration of therapeutic and diagnostic substances at the pathological site. Also, the concentration and dynamics of the active molecule can be varied by controlling the particle size of nanoparticles (>3–5 nm). This control in particle size in conjugation with surface coating with stealth ligand allows them to veil against body's immune system, enabling them to circulate in the blood for longer period of time.^[7] These advances in the field of biotechnology have opened an endless opportunities for molecular diagnostics and therapy.^[8] Once targeted (active or passive), these nanocarriers can be designed in a way to facilitate them to act as imaging probes using variety to techniques such as ultrasound (US), X-ray, computed tomography (CT), positron emission tomography (PET), magnetic resonance imaging (MRI), optical imaging, and surface-enhanced Raman imaging (SERS) [Table 1].^[9] Hence, these so-called “molecular imaging probes” can noninvasively provide valuable information about differentiate abnormalities in various body structures and organs to determine the extent of disease, and evaluate the effectiveness of treatment.^[7] Thus short molecular imaging enables the visualization of the cellular function and the follow-up of the molecular process in living organisms without perturbing them.^[10]

Over the year's nanoparticles such as magnetic nanoparticles (iron oxide), gold and silver nanoparticles, nanoshells, and nanocages have been continuously used and modified to enable their use as a diagnostic and therapeutic agent. Thus, in this particular review article we have introduced iron oxide, gold, and silver nanoparticles along with newer nanoshells and nanocages. These are then briefly discussed for their method

of development and some citing recent examples which utilize their intrinsic properties as diagnostic and/or therapeutic agents for diseases, mainly cancer.

Iron Oxide Nanoparticles

Iron (III) oxide (Fe_2O_3) is a reddish brown, inorganic compound which is paramagnetic in nature and also one of the three main oxides of iron, while other two being FeO and Fe_3O_4 . The Fe_3O_4 , which also occurs naturally as the mineral magnetite, is also superparamagnetic in nature. Due to their ultrafine size, magnetic properties, and biocompatibility, superparamagnetic iron oxide nanoparticles (SPION) have emerged as promising candidates for various biomedical applications, such as enhanced resolution contrast agents for MRI, targeted drug delivery and imaging, hyperthermia, gene therapy, stem cell tracking, molecular/cellular tracking, magnetic separation technologies (e.g., rapid DNA sequencing) early detection of inflammatory, cancer, diabetes, and atherosclerosis.^[11–20] All these biomedical applications require that the nanoparticles have high magnetization values so as to provide high-resolution MR images. In general, the superparamagnetic nanoparticles resemble excellent imaging probes to be used as MRI contrast agents since the MR signal intensity is significantly modulated without any compromise in its *in vivo* stability.^[21] Basically, all contrast agents induce a decrease in the T1 and T2 relaxation times of surrounding water protons and thereby manipulate the signal intensity of the imaged tissue.^[22]

Converging advances in the understanding of the molecular biology of various diseases recommended the need of homogeneous and targeted imaging probes along with a narrow size distribution in between 10 and 250 nm in diameter. Developing magnetic nanoparticles in this diameter range is a

Table 1: Comparison of common imaging techniques along with the nanoparticles currently used or under clinical trials^[10]

Technique	Advantages	Disadvantages	Nanoparticles used
Ultrasound	Easy to perform Noninvasive No radiation hazard because it has nonionizing radiation being emitted Relatively inexpensive as compared to the other imaging modalities currently available	Resolution of images is often limited Reflected very strongly on passing from tissue to gas, or vice versa Does not pass well through bone Attenuation can reduce the resolution of the image	Fe_2O_3 , Gd_2O_3
CT	Wide field of view, Detection of even subtle differences between body tissues Ability to provide cross sectional images of the body	Need for contrast agents for enhanced soft tissue contrast Radiations Tissue non-specificity Cost	Gold and silver nanoparticles Multimodal imaging nanoparticles
PET	Can image biochemical and physiological phenomena	Radiations Some tumors show poor FDG affinity FDG uptake is not limited to the tumor cells and is prevalent in other benign cells Motion artifact is the serious problems Resolution of images is lower as compared to CT or MRI. This results in poor localization of lesions Interpretation is very challenging Most expensive technique	Radioactive ^{64}Cu , ^{62}Cu , ^{82}Rb , and ^{68}Ga , with particles tagged or conjugated with any organic moiety containing ^{19}F being with the first choice
MRI	Higher resolution Can show the anatomical details Does not use any ionizing radiation	Expensive to use Cannot be used in patients with metallic devices, like pacemakers	Iron oxide nanoparticles are most commonly used, but newer generation multimodal imaging agents are also considered for clinical trials

complex process and various chemical routes for their synthesis have been proposed. These methods include microemulsions, sol–gel syntheses, sonochemical reactions, hydrothermal reactions, hydrolysis and thermolysis of precursors, flow injection syntheses, and electrospray syntheses.^[23-29] However, the most common method for the production of magnetite nanoparticles is the chemical coprecipitation technique of iron salts.^[30-34] The main advantage of the coprecipitation process is that a large amount of nanoparticles can be synthesized but with limited control on size distribution. This is mainly due to that the kinetic factors are controlling the growth of the crystal. Thus the particulate magnetic contrast agents synthesized using these methods include ultrasmall particles of iron oxide (USPIO) (10–40 nm), small particles of iron oxide (SPIO) (60–150 nm). Besides, monocrystalline USPIOs are also called as monocrystalline iron oxide nanoparticles (MIONs), whereas MIONs when cross-linked with dextran they are called crosslinked iron oxide nanoparticles CLIO (10–30 nm).^[35-37] The modification of the dextran coating by carboxylation leads to a shorter clearance half-life in blood.^[38] Hence, ferumoxytol (AMAG Pharmaceuticals), a carboxyalkylated polysaccharide coated iron oxide nanoparticle, is already described as a good first-pass contrast agent but uptake by macrophages is unspecific and too fast to enhance the uptake in macrophage-rich plaques.

In order to improve the cellular uptake, these particles can be modified with a peculiar surface coating so that they can be easily conjugated to drugs, proteins, enzymes, antibodies, or nucleotides and can be directed to an organ, tissue, or tumor [Figure 1].^[7,39] While traditional contrast agents distribute rather nonspecifically, targeted molecular imaging probes based on iron oxide nanoparticles have been developed that specifically target body tissue or cells.^[7,40] For instance, Conroy and coworkers developed (chlorotoxin

(CTX)) a biocompatible iron oxide nanoprobe coated with poly(ethylene glycol) (PEG), which is capable of specifically targeting glioma tumors *via* the surface-bound targeting peptide.^[41] Further, MRI studies showed the preferential accumulation of the nanoprobe within gliomas. In another study, Apopa *et al.* engineered iron oxide nanoparticles that can induce an increase in cell permeability through the production of reactive oxygen species (ROS) and the stabilization of microtubules.^[42] These are the few applications of iron oxide nanoparticles in biomedical imaging.

These studies provide a new insight into the bioreactivity of engineered iron nanoparticles, which can provide potential applications in medical imaging or drug delivery. The further development and modification of the complexes of iron oxide along with dendrimers, polymeric nanoparticles, liposomes, and solid lipid nanoparticles are widely studied. However, the toxicity of these magnetic nanoparticles to certain types of neuronal cells is still the matter of concern.^[43]

Gold Nanoparticles

Colloidal gold, also known as gold nanoparticles, is a suspension (or colloid) of nanometer-sized particles of gold. The history of these colloidal solutions dates back to Roman times when they were used to stain glass for decorative purposes.^[44] However, the modern scientific evaluation of colloidal gold did not begin until Michael Faraday's work of the 1850s, when he observed that the colloidal gold solutions have properties that differ from the bulk gold.^[45,46] Hence the colloidal solution is either an intense red color (for particles less than 100 nm) or a dirty yellowish color (for larger particles) as shown in Figures 2 and 3.^[47,48] These interesting optical properties of these gold nanoparticles are due to their unique interaction with light.^[49] In the presence

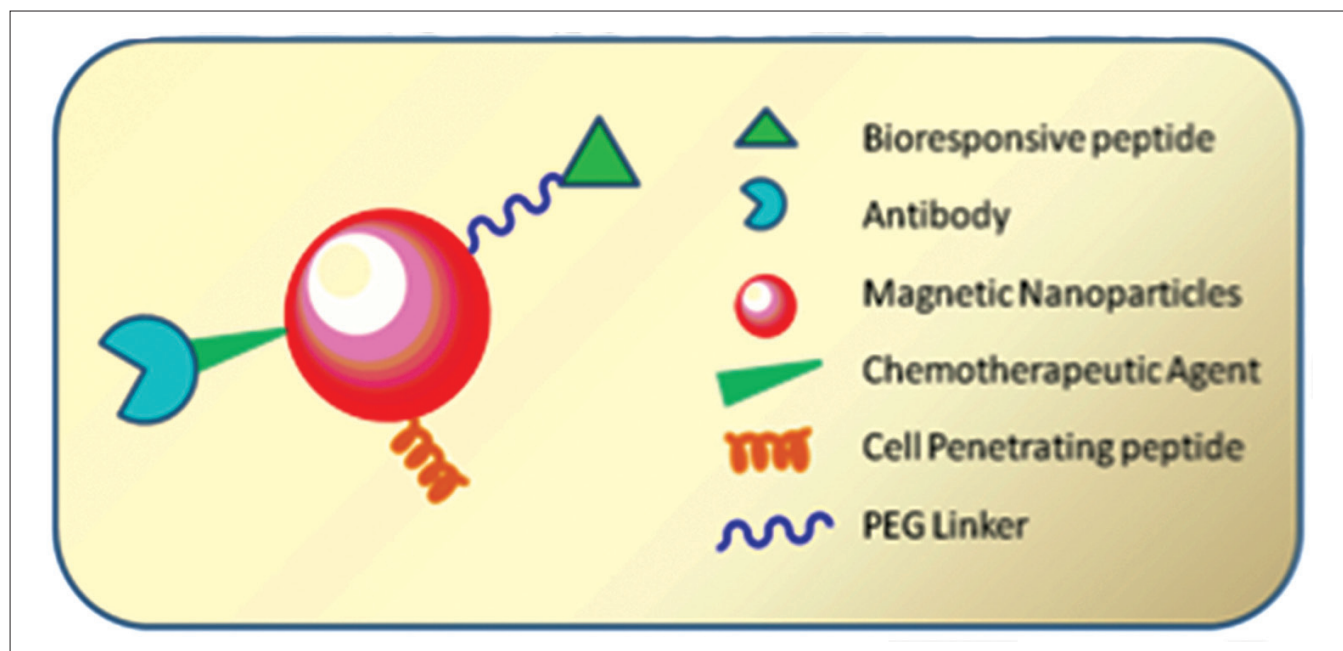


Figure 1: Schematic diagram representing the functionalization of magnetic nanoparticles with bioresponsive peptide, PEG linker, chemotherapeutic agent, antibody, and cell-penetrating peptide.^[11]

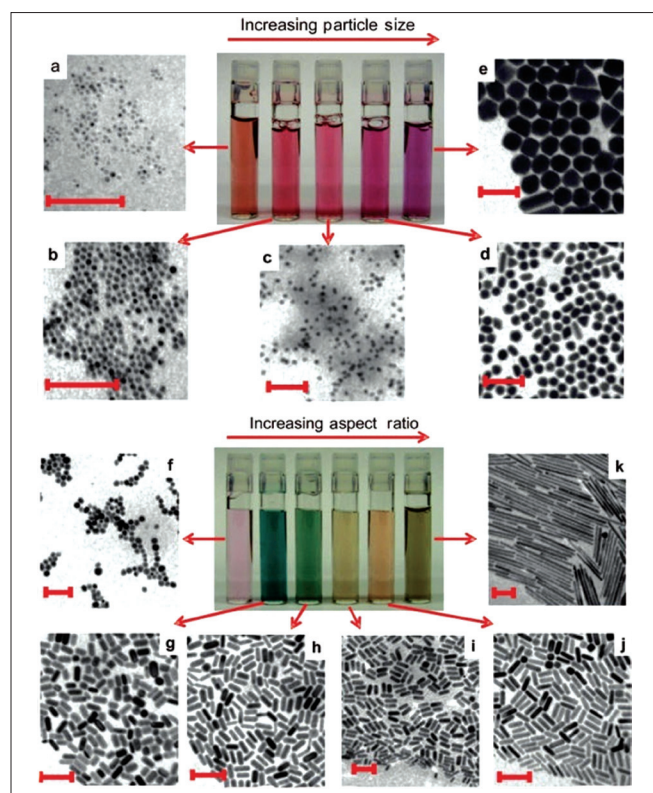


Figure 2: Photographs of aqueous solutions of gold nanospheres (upper panels) and gold nanorods (lower panels) as a function of increasing dimensions. Corresponding transmission electron microscopy images of the nanoparticles are shown (all scale bars 100 nm). The difference in color of the particle solutions is more dramatic for rods than for spheres. This is due to the nature of plasmon bands (one for spheres and two for rods) that are more sensitive to size for rods compared with spheres. For spheres, the size varies from 4 to 40 nm (TEMs a-e), whereas for rods, the aspect ratio varies from 1.3 to 5 for short rods (TEMs f-j) and 20 (TEM k) for long rods^[50]

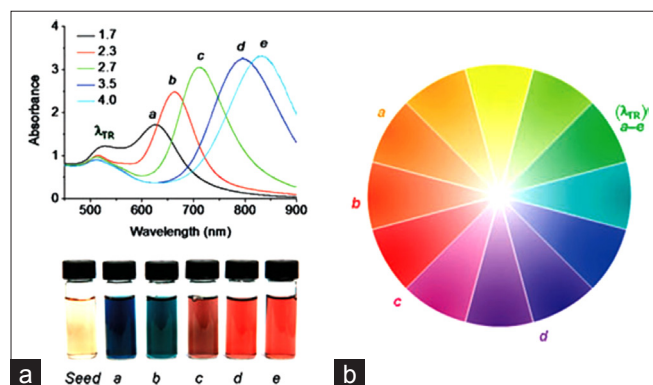


Figure 3: Gold nanorods (NRs) with tunable optical absorptions at visible and near-infrared wavelengths; a) Optical absorption spectra of gold NRs with different aspect ratios (a-e); b) Color wheel, with reference to gold NRs labeled a-e. TR, transverse resonance^[51]

of the oscillating electromagnetic field of the light, the free electrons of the metal nanoparticles undergo an oscillation with respect to the metal lattice.^[50-53] This process is resonant at a particular frequency of the light and is termed the localized surface plasmon resonance (LSPR). After absorption, the

surface plasmon decays radiatively resulting in light scattering or nonradiatively by converting the absorbed light into heat. Thus for gold nanospheres with particle size around 10 nm in diameter have a strong absorption maximum around 520 nm in aqueous solution due to their LSPR. These nanospheres show a Stokes shift with an increase in the nanosphere size due to the electromagnetic retardation in larger particles.

Moreover, the properties and applications of colloidal gold nanoparticles also depend upon its shape. Figure 2 shows that the difference in color of the particle solutions is more dramatic for rods than for spheres. For example, the rod-shaped nanoparticles have two resonances: one due to plasmon oscillation along the nanorod short axis and another due to plasmon oscillation along the long axis, which depends strongly on the nanorod aspect ratio, that is, length-to-width ratio.^[54,55] When the nanorod aspect ratio is increased, the long-axis LSPR wavelength position red shifts from the visible to the NIR and also progressively increases in oscillator strength [Figure 2].^[55] For example, rodlike particles have both transverse and longitudinal absorption peak, and anisotropy of the shape affects their self-assembly.^[56] Due to these unique optical properties, gold nanoparticles are the subject of substantial research, with enormous applications including biological imaging, electronics, and materials science.^[57] Thus to develop gold nanoparticles for specific applications, reliable and high-yielding methods including those with spherical and nonspherical shapes have been developed over the period of years.^[56,58]

The most prevalent method for the synthesis of monodisperse spherical gold nanoparticles was pioneered by Turkevich *et al.* in 1951 and later refined by Frens *et al.* in 1973.^[59-62] This method uses the chemical reduction of gold salts such as hydrogen tetrachloroaurate (HAuCl_4) using citrate as the reducing agent. This method produces monodisperse spherical gold nanoparticles in the range of 10–20 nm in diameter. However, the synthesis of larger gold nanoparticles with diameters between 30 and 100 nm was reported by Brown and Natan *via* seeding of Au^{3+} by hydroxylamine.^[63] Subsequent research led to the modification of the shape of these gold nanoparticles resulting in rod, triangular, polygonal rods, and spherical particles.^[64-66] These ensuing gold nanoparticles have unique properties, providing a high surface area to volume ratio. Moreover, the gold surface offers a unique opportunity to conjugate ligands such as oligonucleotides, proteins, and antibodies containing functional groups such as thiols, mercaptans, phosphines, and amines, which demonstrates a strong affinity for gold surface.^[67] The realization of such gold nanoconjugates coupled with strongly enhanced LSPR gold nanoparticles have found applications in simpler but much powerful imaging techniques such as dark-field imaging, SERS, and optical imaging for the diagnosis of various disease states.^[68]

In fact, El Sayed *et al.* have established the use of gold nanoparticles for cancer imaging by selectively transporting AuNPs into the cancer cell nucleus. In order to selectively transport the AuNPs into the cancer cell nucleus, they conjugated arginine–glycine–aspartic acid peptide (RGD) and a nuclear localization signal peptide (NLS) to a 30-nm AuNPs *via* PEG.^[69]

RGD is known to target $\alpha_v\beta_3$ integrins receptors on the surface of the cell, whereas NLS sequence lysine–lysine–lysine–arginine–lysine (KKKRK) sequence is known to associate with karyopherins (importins) in the cytoplasm, which enables the translocation to the nucleus.^[70-72] Thus the presence of RGD will enable cancer-cell-specific targeting, whereas the presence of NLS will exhibit cancer cell nucleus specific targeting. This intuitively developed particle was then targeted to human oral squamous cell carcinoma (HSC) having $\alpha_v\beta_3$ integrins overexpressed on the cell surface (cancer model), and human keratinocytes (HaCat) (control). The authors further demonstrated that RGD-AuNPs specifically target the cytoplasm of cancer cells [Figure 4a] over that of normal cells [Figure 4c], and the RGD/NLS-AuNPs specifically target the nuclei of cancer cells [Figure 4b] over those of normal cells [Figure 4d].^[69]

Similarly, Qian *et al.* reported the development of tumor-targeted gold nanoparticles as a probe for Raman scatters *in vivo*.^[73] These gold nanoparticles were encoded with a Raman reporter and further encapsulated with a thiol-modified PEG coat. Additionally, to specifically target tumor cells, the pegylated gold nanoparticles were then conjugated with an antibody against epidermal growth factor receptor, which is sometimes overexpressed in certain types of cancer cells. The Raman enhancement from these tailored particles was then observed with electronic transitions at 633 or 785 nm *via* SERS.

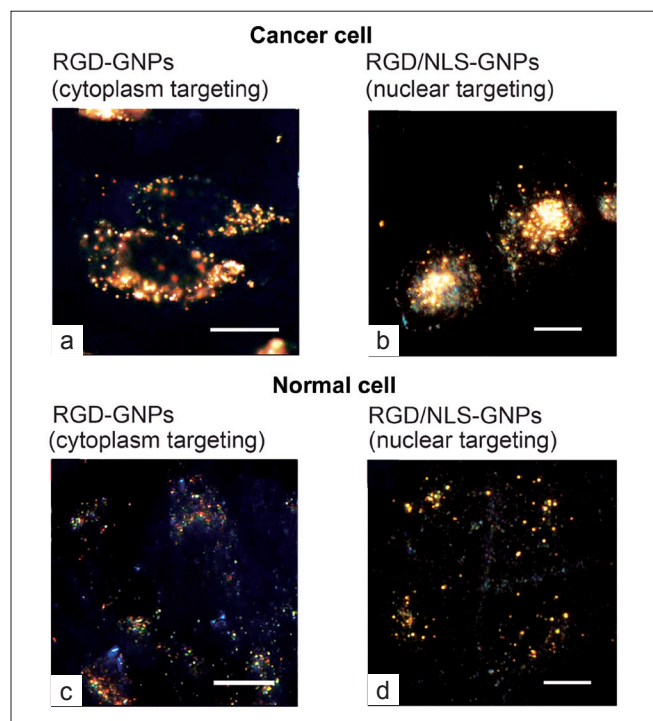


Figure 4: Dark field light scattering images of cytoplasm and nuclear targeting AuNPs. a) RGD-AuNPs located in the cytoplasm of cancer cells. b) RGD/NLS-AuNPs located at the nucleus of cancer cells. c) RGD-AuNPs located in the cytoplasm of normal cells. d) RGD/NLS-AuNPs located at the nucleus of normal cells. The cancer and normal cells were incubated in the presence of these AuNPs at a concentration of 0.4 nM for 24 hours and these images clearly display the efficient uptake of AuNPs in cancer cells compared with normal cells. Scale bar 10 μm ^[72]

The results obtained by Qian and coworkers suggest the highly specific recognition and detection of human cancer cells, as well as active targeting of EGFR-positive tumor xenografts in animal models can be made using SERS.^[73]

Moreover, the use of gold nanorods as photothermal agents sets them apart from all nanoprobe. Photothermal therapy (PTT) is a procedure in which a photosensitizer is excited with specific band light (mainly IR). This activation brings the sensitizer to an excited state where it then releases vibrational energy in the form of heat. The heat is the actual method of therapy that kills the targeted cells. One of the biggest recent successes in photothermal therapy is the use of gold nanoparticles. Spherical gold nanoparticles absorptions have not been optimal for *in vivo* applications. This is because the peak absorptions have been limited to 520 nm for 10 nm diameter. Moreover, skin, tissues, and hemoglobin have a transmission window from 650 up to 900 nm. This was circumvented by the recent invention of gold nanorods by Murphy and Coworkers, who were able to tune the absorption peak of these nanoparticles, which can also be tuned from 550 nm up to 1 μm just by altering its aspect ratio of the nanorods [Figure 3].^[48,74] Hence, for the rod-shaped gold nanoparticles with the absorption in the IR region, when selectively accumulated in tumors when bathed in laser light (in the IR region), the surrounding tissue is barely warmed, but the nanorods convert light to heat, killing the malignant cells. This potential application of gold nanorods sanctifies them from other nanoprobe. However, their incompatibility with other high-resolution imaging techniques such as MRI and irreproducibility in shapes led to the invention of nanocages and nanoshells.

Nanoshells and Nanocages

Neeves and Birnboim calculated that a composite spherical particle consisting of a metallic shell and a dielectric core could give rise to LSPR modes with their wavelengths tunable over a broad range of the electromagnetic spectrum.^[75] Later on, the experimental and theoretical work by Naomi Halas and Peter Nordlander at Rice University showed that the resonance of a silica–gold nanoshell particle can easily be positioned in the near-infrared (800–1,300 nm) region, where absorption by biomatters is low [Figure 5].^[76-78] They developed silica–gold nanospheres by using freshly formed amine-terminated silica spheres. These amine terminated silica spheres were then treated with a suspension of gold colloids (1–2 nm in size). Gold was deposited *via* chemical reduction to cover the silica core and to the amine terminal of the silicon core. Although this method is widely used, the intricacy involved in the control of thickness and smoothness of the metallic shells makes this method unsuitable for the routine synthesis of controlled particle-sized nanoshells. Furthermore, they also showed the successful irreversible photothermal ablation of tumor tissue both *in vitro* and *in vivo* when these nanoshells were localized onto the tumor cells. In another study, Halas and West established the use of near-infrared resonant nanoshells for whole-blood immunoassays. They further showed that the nanoshells when conjugated with antibodies act as recognition sites for a specific analyte [Figure 6]. The analyte causes

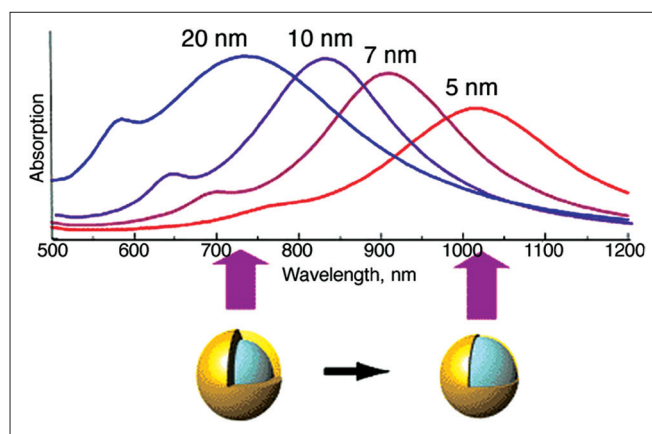


Figure 5: Gold nanoshell plasmon resonances for a 120-nm core with indicated shell thickness^[81]

the formation of dimmers, which will modify the LSPR.^[79] Subsequent work in this field led to the development of the multifunctional magnetic gold nanoshells (Mag-GNS) by Jaeyun *et al.* utilizing Fe_3O_4 nanoparticles as the magnetic core. The Fe_3O_4 nanoparticles allow MRI for diagnosis, and the gold nanoshells enable photothermal therapy. By attaching an antibody to the Mag-GNS by a PEG linker, cancer cells can be targeted. Once localized, these particles enable the detection of cancer using MRI, whereas the photothermal therapy can be used to get rid of cancer cells.^[80]

Similar to gold nanoshells, gold nanocages represent a novel class of nanostructures that are hollow porous gold nanoparticles that absorb light in the near-infrared range. They were first developed by the Xia and Coworkers *via* the reaction of silver nanoparticles with chloroauric acid (HAuCl_4) in boiling water.^[81] Their LSPR peaks can also be tuned to the near infrared region by controlling the thickness and porosity of the walls. Comparable to nanoshells they have found applications in drug delivery and/or controlled drug release. Furthermore, the hollow interiors can host small objects such as magnetic nanoparticles to construct multifunctional hybrid nanostructures diagnostic imaging and therapy.

Silver Nanoparticles

Silver nanoparticles are the particles of silver, with particle size between 1 and 100 nm in size. While frequently described as being “silver” some are composed of a large percentage of silver oxide due to their large ratio of surface to bulk silver atoms. Like gold nanoparticles, ionic silver has a long history and was initially used to stain the glass for yellow. Currently, there is also an effort to incorporate silver nanoparticles into a wide range of medical devices, including bone cement, surgical instruments, surgical masks, etc. Moreover, it has also been shown that ionic silver, in the right quantities, is suitable in treating wounds.^[82-84]

In fact, silver nanoparticles are now replacing silver sulfadiazine as an effective agent in the treatment of wounds. Additionally, Samsung has created and marketed a material called Silver Nano, which includes silver nanoparticles on the surfaces of household appliances. Moreover, due to their attractive

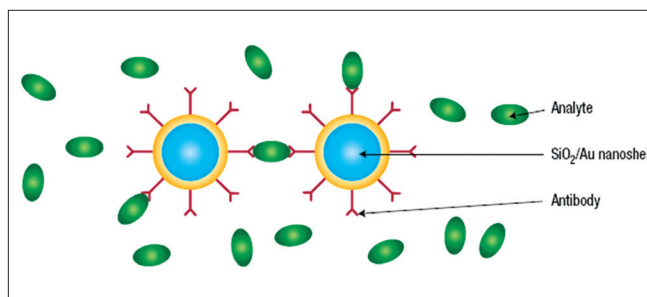


Figure 6: Formation of nanoshell dimer with the interaction of antibodies immobilized on the surface of the nanoshells^[82]

physiochemical properties these nanomaterials have received considerable attention in biomedical imaging using SERS. In fact, the surface plasmon resonance and large effective scattering cross-section of individual silver nanoparticles make them ideal candidates for molecular labeling.^[85] Thus many targeted silver oxide nanoprobe are currently being developed.

Typically, they are synthesized by the reduction of a silver salt with a reducing agent like sodium borohydride in the presence of a colloidal stabilizer. The most common colloidal stabilizers used are polyvinyl alcohol, poly(vinylpyrrolidone), bovine serum albumin (BSA), citrate, and cellulose. Newer novel methods include the use of β -D-glucose as a reducing sugar and a starch as the stabilizer to develop silver nanoparticles ion implantation used to create silver nanoparticles.^[86] Also, it is important to note that not all nanoparticles created are equal. The size and shape have been shown to have an impact on its efficacy. In fact, Elechiguerra *et al.* demonstrated that silver nanoparticles undergo a size-dependent interaction with HIV-1, with particles exclusively in the range of 1–10 nm attached to the virus.^[87] They further suggest that silver nanoparticles interact with the HIV-1 virus via preferential binding to the gp120 glycoprotein knobs.^[87] Similarly, Furno and Coworkers have developed biomaterials by impregnating silicone coated with silver oxide nanoparticles using supercritical carbon dioxide.^[88] These novel biomaterials were developed with an aim to reduce the antibacterial infection. The results obtained were mixed but the methodology allows for the first-time silver impregnation (as opposed to coating) of medical polymers and promises to lead to an antimicrobial biomaterial.^[88]

Even though these particles are not as widely preferred as compared to the gold nanoparticles and nanoshells, but they have made a tremendous impact on today's era of medical science. The interesting property of the noble metals is a promise that they would be continuously used as newer applications and protocols are being developed.

Conclusion

This review article provides a glimpse to some simpler nanoparticles which are being currently modified for their potential applications in medicine. However, the field of nanoscience has blossomed over the last two decades and the need for nanotechnology to explore beyond the cells walls has

become more important.^[89] Nanoparticles have successfully come to aid various disease states, but the advances in biomedical imaging depend largely on the shape, size, and selectivity of the nanoparticle to the target.^[89] Moreover, the type of the particle synthesized also governs the imaging modality to be used and thus the cost of diagnosis. Even though current investigations have demonstrated that multivalent composite materials can provide significant advantages, the ambiguity in developing them for a particular target with high specificity is still challenging. Fortunately, the field of nanotechnology continues to grow interest within the chemical research community with major discoveries as well as new scientific challenges.^[44] Nevertheless, the future studies should also aim to address safety and biocompatibility of these nanoparticles, in particular long-term toxicities.^[90] Additional clinical studies on humans and on animal models should be performed to substantiate their use especially in biomedical imaging using MRI, CT, ultrasound, PET, SERS, and optical imaging.^[90]

References

- Salata O. Applications of nanoparticles in biology and medicine. *J Nanobiotechnol* 2004;2:3.
- Kang YS, Risbud S, Rabolt JF, Stroeve P. Synthesis and characterization of nanometer-size Fe₃O₄ and γ-Fe₂O₃ Particles. *Chem Mater* 1996;8:2209-11.
- Pankhurst QA, J Connolly, Jones SK, Dobson J. Applications of magnetic nanoparticles in biomedicine. *J Phys D Appl Phys* 2003;36:R167.
- Dobson J. Gene therapy progress and prospects: magnetic nanoparticle-based gene delivery. *Gene Ther* 2006;13:283-7.
- Rudge S, Peterson C, Vessely C, Koda J, Stevens S, Catterall L. Adsorption and desorption of chemotherapeutic drugs from a magnetically targeted carrier (MTC). *J Control Release* 2001;74:335-40.
- Appenzeller T. The man who dared to think small. *Science* 1991;254:1300.
- Mody VV, Nounou MI, Bikram M. Novel nanomedicine-based MRI contrast agents for gynecological malignancies. *Adv Drug Deliv Rev* 2009;61:795-807.
- Praetorius NP, Mandal TK. Mandal, Engineered nanoparticles in cancer therapy. *Recent Pat Drug Deliv Formul* 2007;1:37-51.
- Sharma P, Brown S, Walter G, Santra S, Moudgil B. Nanoparticles for bioimaging. *Adv Colloid Interface Sci* 2006;123-126:471-85.
- Moghim SM, Hunter AC, Murray JC. Nanomedicine: current status and future prospects. *FASEB J* 2005;19:311-30.
- Morales MP, Bomati-Miguel O, de Alejo RP, Ruiz-Cabello J, Veintemillas-Verdaguer S, O'Grady K. Contrast agents for MRI based on iron oxide nanoparticles prepared by laser pyrolysis. *J Magn Magn Mater* 2003;266:102-9.
- Babes L, Denizot B, Tanguy G, Le Jeune JJ, Jallet P. Synthesis of iron oxide nanoparticles used as MRI contrast agents: A Parametric Study. *J Colloid Interface Sci* 1999;212:474-82.
- Yigit MV, Mazumdar D, Lu Y. MRI detection of thrombin with aptamer functionalized superparamagnetic iron oxide nanoparticles. *Bioconjug Chem* 2008;19:412-7.
- Peng XH, Qian X, Mao H, Wang AY, Chen ZG, Nie S. Targeted magnetic iron oxide nanoparticles for tumor imaging and therapy. *Int J Nanomed* 2008;3:311-21.
- Chertok B, Moffat BA, David AE, Yu F, Bergemann C, Ross BD. Iron oxide nanoparticles as a drug delivery vehicle for MRI monitored magnetic targeting of brain tumors. *Biomaterials* 2008;29:487-96.
- Weimuller G, Zeisberger M, Krishnan KM. Size-dependant heating rates of iron oxide nanoparticles for magnetic fluid hyperthermia. *J Magn Magn Mater* 2009;321:1947-50.
- Wei W, Xu C, Wu H. Magnetic iron oxide nanoparticles mediated gene therapy for breast cancer - an *in vitro* study. *J Huazhong Univ Sci Technolog Med Sci* 2006;26:728-30.
- Elias A, Tsourkas A. Imaging circulating cells and lymphoid tissues with iron oxide nanoparticles. *Hematology* 2009;2009:720-6.
- Moore A, Marecos E, Bogdanov A Jr, Weissleder R. Tumoral distribution of long-circulating dextran-coated iron oxide nanoparticles in a rodent model. *Radiology* 2000;214:568-74.
- Kooi ME, Cappendijk VC, Cleutjens KB, Kessels AG, Kitslaar PJ, Borgers M. Accumulation of ultras small superparamagnetic particles of iron oxide in human atherosclerotic plaques can be detected by *in vivo* magnetic resonance imaging. *Circulation* 2003;107:2453-8.
- Hildebrandt N, Hermsdorf D, Signorelli R, Schmitz SA, Diederichsen U. Superparamagnetic iron oxide nanoparticles functionalized with peptides by electrostatic interactions. *ARKIVOC* 2007;5:79-90.
- Mahfouz AE, Hamm B, Taupitz M. Contrast agents for MR imaging of the liver: a clinical overview. *Eur Radiol* 1997;7:507-13.
- Chin AB, Yaacob II. Synthesis and Characterization of magnetic Iron Oxide Nanoparticles via microemulsion and Massart's procedure *J Mater Process Technol* 2007;191:235.
- Albornoz C, Jacobo SE. Preparation of a biocompatible magnetic film from an aqueous ferrofluid. *J Magn Magn Mater* 2006;305:12.
- Kim EH, Lee HS, Kwak BK, Kim BK. Synthesis of ferrofluid with magnetic nanoparticles by sonochemical method for MRI contrast agent. *J Magn Magn Mater* 2005;289:328.
- Wan J, Chen X, Wang Z, Yang X, Qian Y. A soft-template-assisted hydrothermal approach to single-crystal Fe₃O₄ nanorods. *J Cryst Growth* 2005;276:571.
- Kimata M, Nakagawa D, Hasegawa M. Preparation of monodisperse magnetic particles by hydrolysis of iron alkoxide. *Powder Technol* 2003;132:112.
- Alvarez GS, Muhammed M, Zagorodni AA. Novel flow injection synthesis of iron oxide nanoparticles with narrow size distribution. *Chem Eng Sci* 2006;61:4625.
- Basak S, Chen DR, Biswas P. Electrospray of ionic precursor solutions to synthesize iron oxide nanoparticles: Modified scaling law. *Chem Eng Sci* 2007;62:1263.
- Martinez-Mera I, Espinosa ME, Perez-Hernandez R, Arenas-Alatorre. Synthesis of magnetite (Fe₃O₄) nanoparticles without surfactants at room temperature. *J Mater Lett* 2007;61:4447.
- Morrison SA, Cahill CL, Carpenter EE, Calvin S, Harris VG. Atomic engineering of mixed ferrite and core-shell nanoparticles. *J Nanosci Nanotechnol* 2005;5:1323-44.
- Sun S, Zeng H, Robinson DB, Raoux S, Rice PM, Wang SX, Li G. Monodisperse MFe₂O₄ (M = Fe, Co, Mn) nanoparticles. *J Am Chem Soc* 2004;126:273-9.
- Qiu J, Yang R, Li M, Jiang N. Preparation and characterization of porous ultrafine Fe₃O₄ particles. *Mater Res Bull* 2005;40:1968.
- Lee SJ, Jeong JR, Shin SC, Kim JC, Kim JD. Synthesis and characterization of superparamagnetic maghemite nanoparticles prepared by coprecipitation technique. *J Magn Magn Mater* 2004;282:147.
- Shen T, Weissleder R, Papisov M, Bogdanov A Jr, Brady TJ. Monocrystalline iron oxide nanocompounds (MION): physicochemical properties. *Magn Reson Med* 1993;29:599-604.
- Weissleder R, Lee AS, Khaw BA, Shen T, Brady TJ. Antimyosin-labeled monocrystalline iron oxide allows detection of myocardial infarct: MR antibody imaging. *Radiology* 1992;182:381-5.
- Montet X, Weissleder R, Josephson L. Imaging pancreatic cancer with a peptide-nanoparticle conjugate targeted to normal pancreas. *Bioconjugate Chem* 2006;17:905-11.
- Hasegawa M, Hanaichi T, Shoji H, Kawaguchi T, Maruno S. Biological behaviour of Dextran-Iron oxide injected intravenously in rats. *Jpn J Appl Phys* 1998;37:1029.
- Chastellain M, Petri, A Gupta A, Rao KV, Hofmann H. Scalable synthesis of new class of polymer microrods by a liquid-liquid dispersion technique. *Adv Eng Mater* 2004;6:235.
- Mazooz G, Mehlman T, Lai TS, Greenberg CS, Dewhirst MW, Neeman M. Development of magnetic resonance imaging contrast material for *in vivo* mapping of tissue transglutaminase activity. *Cancer Res* 2005;65:1369-75.
- Sun C, Veisheh O, Gunn J, Fang C, Hansen S, Lee D, et al. *In vivo* MRI detection of gliomas by chlorotoxin-conjugated superparamagnetic nanoparticles. *Small* 2008;4:372-9.
- Apopa PL, Qian Y, Shao R, Guo NL, Schwegler-Berry D, Pacurari M, et al. Iron oxide nanoparticles induce human microvascular endothelial cell permeability through reactive oxygen species production and microtubule remodeling. *Part Fibre Toxicol* 2009;6:1.
- Pisanic TR 2nd, Blackwell JD, Shubayev VI, Fiñones RR, Jin S. Nanotoxicity of iron oxide nanoparticle internalization in growing

- neurons. *Biomaterials* 2007;28:2572-81.
44. Giljohann DA, Seferos DS, Daniel WL, Massich MD, Patel PC, Mirkin CA. Gold nanoparticles for biology and medicine. *Angew Chem Int Ed Engl* 2010;49:3280-94.
 45. Peter PE, Meurig JT. *Fein verteiltes Gold - Faradays Beitrag zu den heutigen Nanowissenschaften* 13. *Angewandte Chemie* 2007;119:5576-82.
 46. Hayat MA. *Colloidal gold: principles, methods, and applications*. San Diego: Academic Press; 1989.
 47. Murphy CJ, Gole AM, Stone JW, Sisco PN, Alkilany AM, Goldsmith EC, *et al.* Gold nanoparticles in biology: beyond toxicity to cellular imaging. *Acc Chem Res* 2008;41:1721-30.
 48. Tong L, Wei Q, Wei A, Cheng JX. Gold nanorods as contrast agents for biological imaging: optical properties, surface conjugation and photothermal effects. *Photochem Photobiol* 2009;85:21-32.
 49. Jain PK, Huang X, El-Sayed IH, El-Sayed MA. Noble metals on the nanoscale: optical and photothermal properties and some applications in imaging, sensing, biology, and medicine. *Acc Chem Res* 2008;41:1578-86.
 50. Jain PK, Lee KS, El-Sayed IH, El-Sayed MA. Calculated absorption and scattering properties of gold nanoparticles of different size, shape, and composition: applications in biological imaging and biomedicine. *J Phys Chem B* 2006;110:7238-48.
 51. Kelly KL, Coronado E, Zhao LL, Schatz GC. The optical properties of metal nanoparticles: the influence of size, shape, and dielectric environment. *J Phys Chem B* 2002;107:668-77.
 52. Kreibitz U, Vollmer M. *Optical Properties of Metal Clusters*. Springer Series Materials Sci 1995. p. 25.
 53. Link S, El-Sayed MA. Optical properties and ultrafast dynamics of metallic nanocrystals. *Annu Rev Phys Chem* 2003;54:331-66.
 54. Murphy CJ, Sau TK, Gole AM, Orendorff CJ, Gao J, Gou L, *et al.* Anisotropic metal nanoparticles: Synthesis, assembly, and optical applications. *J Phys Chem B* 2005;109:13857-70.
 55. Link S, Mohamed MB, El-Sayed MA. Simulation of the optical absorption spectra of a gold nanorods as a function of their aspect ratio and the effect of the medium dielectric constant. *J Phys Chem B* 1999;103:3073.
 56. Sharma V, Park K, Srinivasarao M. Colloidal dispersion of gold nanorods: Historical background, optical properties, seed-mediated synthesis, shape separation and self-assembly. *Mater Sci Eng R Rep* 2009;65:1-38.
 57. Rao RC, Kulkarni GU, Thomas JP, Edwards PP. Metal nanoparticles and their assemblies. *Chem Soc Rev* 2000;29:27-35.
 58. Daniel MC, Astruc D. Gold nanoparticles: assembly, supramolecular chemistry, quantum-size-related properties, and applications toward biology, catalysis, and nanotechnology. *Chem Rev* 2004;104:293-346.
 59. Turkevich J, Stevenson PC, Hillier J. A study of the nucleation and growth processes in the synthesis of colloidal gold. *Discuss Faraday Soc* 1951;11:55-75.
 60. Kimling J, Maier M, Okenve B, Kotaidis V, Ballot H, Plech A. turkevich method for gold nanoparticle synthesis revisited. *J Phys Chem B* 2006;110:15700-7.
 61. Frens G. Particle size and sol stability in metal colloids. *Colloid Polym Sci* 1972;250:736-41.
 62. Frens G. Controlled nucleation for regulation of particle size in monodisperse gold suspensions. *Na Phys Sci* 1973;241:20-2.
 63. Brown KR, Natan MJ. Hydroxylamine seeding of colloidal Au nanoparticles in solution and on surfaces. *Langmuir* 1998;14:726-8.
 64. Jana NR, Gearheart L, Murphy CJ. Seed-mediated growth approach for shape-controlled synthesis of spheroidal and rod-like gold nanoparticles using a surfactant template. *Adv Mater* 2001;13:1389-93.
 65. Jana NR, Gearheart L, Murphy CJ. Wet chemical synthesis of high aspect ratio cylindrical gold nanorods. *J Phys Chem B* 2001;105:4065-7.
 66. Subrata K, Sudipa P, Snigdhmayee P, Soumen B, Sujit KG, Anjali P, *et al.* Anisotropic growth of gold clusters to gold nanocubes under UV irradiation. *Nanotechnology* 2007;18:075712.
 67. Alivisatos AP, Johnsson KP, Peng X, Wilson TE, Loweth CJ, Bruchez MP Jr. Organization of 'nanocrystal molecules' using DNA. *Nature* 1996;382:609-11.
 68. El-Sayed IH, Huang X, El-Sayed MA. Surface plasmon resonance scattering and absorption of anti-EGFR antibody conjugated gold nanoparticles in cancer diagnostics: applications in oral cancer. *Nano Lett* 2005;5:829.
 69. Kang B, Mackey MA, El-Sayed MA. Nuclear targeting of gold nanoparticles in cancer cells induces DNA damage, causing cytokinesis arrest and apoptosis. *J Am Chem Soc* 2010;132:1517-9.
 70. Gao H, Shi W, Freund LB. Mechanics of receptor-mediated endocytosis. *Proc Natl Acad Sci U S A* 2005;102:9469-74.
 71. Nakiely S, Dreyfuss G. Transport of Proteins and RNAs in and out of the nucleus. *Cell* 1999;99:677-90.
 72. Xue H, Atakilit A, Zhu W, Li X, Ramos DM, Pytela R. Role of the [alpha]v[beta]6 integrin in human oral squamous cell carcinoma growth in vivo and in vitro. *Biochem Biophys Res Commun* 2001;288:610-8.
 73. Qian X, Peng XH, Ansari DO, Yin-Goen Q, Chen GZ, Shin DM, *et al.* In vivo tumor targeting and spectroscopic detection with surface-enhanced Raman nanoparticle tags. *Nat Biotechnol* 2008;26:83-90.
 74. Busbee BD, Obare SO, Murphy CJ. An improved synthesis of high-aspect-ratio gold nanorods. *Adv Mater* 2003;15:414-6.
 75. Neeves AE, Birnboim MH. Composite structures for the enhancement of nonlinear-optical susceptibility. *J Opt Soc Am B* 1989;6:787-96.
 76. Oldenburg SJ, Averitt RD, Westcott SL, Halas NJ. Nanoengineering of optical resonances. *Chem Phys Lett* 1998;288:243-7.
 77. Prodan E, Lee A, Nordlander P. The effect of a dielectric core and embedding medium on the polarizability of metallic nanoshells. *Chem Phys Lett* 2002;360:325-32.
 78. Hirsch LR, Stafford RJ, Bankson JA, Sershen SR, Rivera B, Price RE, *et al.* Nanoshell-mediated near-infrared thermal therapy of tumors under magnetic resonance guidance. *Proc Natl Acad Sci U S A* 2003;100:13549-54.
 79. Brongersma ML. Nanoscale photonics: Nanoshells: gifts in a gold wrapper. *Nat Mater* 2003;2:296-7.
 80. Kim J, Park S, Lee JE, Jin SM, Lee JH, Lee IS, *et al.* Designed fabrication of multifunctional magnetic gold nanoshells and their application to magnetic resonance imaging and photothermal therapy. *Angew Chem Int Ed Engl* 2006;45:7754-8.
 81. Chen J, Saeki F, Wiley BJ, Cang H, Cobb MJ, Li ZY, *et al.* Gold nanocages: bioconjugation and their potential use as optical imaging contrast agents. *Nano Letters* 2005;5:473-7.
 82. Qin Y. Silver-containing alginate fibres and dressings. *Int Wound J* 2005;2:172-6.
 83. Atiyeh BS, Costagliola M, Hayek SN, Dibo SA. Effect of silver on burn wound infection control and healing: review of the literature. *Burns* 2007;33:139-48.
 84. Lansdown AB. Silver in health care: antimicrobial effects and safety in use. *Curr Probl Dermatol* 2006;33:17-34.
 85. Schultz S, Smith DR, Mock JJ, Schultz DA. Single-target molecule detection with nonbleaching multicolor optical immunolabels. *Proc Natl Acad Sci U S A* 2000;97:996-1001.
 86. Stepanov AL, Popok VN, Hole DE. Formation of Metallic Nanoparticles in Silicate Glass through Ion Implantation. *Glass Phy Chem* 2002;28:90-5.
 87. Elechiguerra JL, Burt JL, Morones JR, Camacho-Bragado A, Gao X, Lara HH, *et al.* Interaction of silver nanoparticles with HIV-1. *J Nanobiotechnol* 2005;3:6.
 88. Furno F, Morley KS, Wong B, Sharp BL, Arnold PL, Howdle SM, *et al.* Silver nanoparticles and polymeric medical devices: a new approach to prevention of infection? *J Antimicrob Chemother*. 2004;54:1019-24.
 89. Evanoff DD Jr, Chumanov G. Synthesis and optical properties of silver nanoparticles and arrays. *Chemphyschem* 2005;6:1221-31.
 90. Laurent S, Forge D, Port M, Roch A, Robic C, Vander Elst L, *et al.* Magnetic iron oxide nanoparticles: synthesis, stabilization, vectorization, physicochemical characterizations, and biological applications. *Chem Rev* 2008;108:2064-110.

Source of Support: Nil, **Conflict of Interest:** None declared.